

Split Iterated-Squaring Polynomials and Norm-like Gaussian Integers - A Research Proposal

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Abstract

We investigate the search for integers $(L_1, L_2, L_3, L_4, L_5)$ such that the polynomial $((((x - L_1)^2 - L_2)^2 - L_3)^2 - L_4)^2 - L_5$ completely factors over the integers. Such a finding will provide an ideal symmetric solution for the Prouhet-Tarry-Escott (PTE) problem with degree $n = 15$, extending the state of the art, currently at $n = 11$. By linking the problem to the study of norm-like Gaussian integers, we reduced the search problem to finding zeros of a single multivariate polynomial of a special form. In addition, we provide a new enumeration technique for possible solutions. We believe that further results can be achieved following the proposed directions, as well as by running extensive enumerations and examining numerical data.

Introduction

The E_n Problem Statement

Given integers $L_1, L_2, \dots, L_n$, we define the associated E_n **Polynomial** recursively as:

$$E_1(x; L) := x^2 - L$$
$$E_n(x; L_1, \dots, L_n) := E_1(E_{n-1}(x; L_1, \dots, L_{n-1}); L_n)$$

The E_n **Problem** is to find integers $(L_1, L_2, \dots, L_n)$ such that the polynomial $E_n(x; L_1, \dots, L_n)$ completely factors over the integers. We refer to such a tuple as an E_n **solution**. We can therefore formulate our primary goal as finding a **non-trivial E_5 solution**. The equivalence of E_n solutions and ideal symmetric solutions for the PTE problems is discussed in [1].

Necessary and Sufficient Conditions for an E_n Solution

It is clear that if L_n is not a perfect square, then E_n polynomial cannot have integer roots. If L_n is a perfect square, we can split E_n polynomial into a product of two E_{n-1} polynomials:

$$E_n(x; L_1, \dots, L_n) = E_n(x; L_1, \dots, L_{n-1} - \sqrt{L_n})E_n(x; L_1, \dots, L_{n-1} + \sqrt{L_n})$$

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We can continue with this reasoning in a binary tree-like fashion until we recover the full decomposition of E_n :

$$E_n(x; L_1, \dots, L_n) = \prod_{\nu=0}^{2^{n-1}-1} (x - r_\nu)(x + r_\nu)$$

$$r_\nu := \sqrt{L_1 \pm \sqrt{L_2 \pm \sqrt{\dots \pm \sqrt{L_n}}}}$$

With $r_0, \dots, r_{2^{n-1}-1}$ being the 2^{n-1} positive roots of E_n and the $\pm$ pattern follows the binary expansion of the index ν . The positive roots are canonically sorted,

and the constraint $\sqrt{L_k \pm \sqrt{L_{k+1} \pm \sqrt{\dots \pm \sqrt{L_n}}}} \in \mathbb{Z}$ must hold at every node of the tree. One can rewrite the constraints in terms of the roots. The lowest three constraints are:

$$\begin{aligned} r_{2k}^2 + r_{2k+1}^2 &= 2L_1 \\ (r_{4k}^2 - r_{4k+1}^2)^2 + (r_{4k+2}^2 - r_{4k+3}^2)^2 &= 8L_2 \\ ((r_{8k}^2 - r_{8k+1}^2)^2 - (r_{8k+2}^2 - r_{8k+3}^2)^2) + ((r_{8k+4}^2 - r_{8k+5}^2)^2 - (r_{8k+6}^2 - r_{8k+7}^2)^2) &= 32L_3 \end{aligned}$$

We refer to a constraint at the n -th level as the L_n **condition**. Note that an $L_{n>1}$ condition possesses certain degrees of freedom arising from the lower conditions in the tree. In [1], an equivalent set of conditions is introduced under the name **Litter Conditions**. Useful augmentations of the L_2 condition include:

$$\begin{aligned} L_{2,a} &:= r_0^4 + r_1^4 + r_2^4 + r_3^4 = 4(L_1^2 + L_2) \\ L_{2,b} &:= r_0^2 r_1^2 + r_2^2 r_3^2 = 2(L_1^2 - L_2) \\ L_{2,c} &:= (r_0^2 - L_1)^2 + (r_2^2 - L_1)^2 = 2L_2 \\ L_{2,d} &:= (r_0^2 - r_1^2)^2 - 4r_0^2 r_1^2 + (r_2^2 - r_3^2)^2 - 4r_2^2 r_3^2 = 8(2L_2 - L_1^2) \end{aligned}$$

Key Property - Solution Construction

If two E_n solutions share identical $L_1, \dots, L_{n-1}$ coefficients, they can be used to construct an E_{n+1} solution. This suggests a strategy for finding E_5 solutions by first finding many E_4 solutions, and then find a pair of solutions that share L_1, L_2 and L_3 .

The Connection to Gaussian Integers

Since the L_1 condition for all root pairs implies $r_{2k}^2 + r_{2k+1}^2 = 2L_1$, we can treat the root pairs as Gaussian integers of equal norm. When searching for roots

that satisfy higher order L_n conditions, we can confine our search to the complex circle of radius $2L_1$. By fixing L_1 and examining its factorization, we can:

1. Determine the number of points with norm $2L_1$ (namely the sum of two squares function, $r_2(n)$ [2]).
2. Efficiently compute the set of all such points. Such a set provides the basis for a search that can yield E_3 , E_4 and higher order solutions.
3. Parameterize sets of norm-like Gaussian integers even without a concrete L_1 at hand.

Factorization Analysis

The following is the standard derivation of the factorization of an arbitrary rational integer over the Gaussian integers.

Let n be a norm factoring as:

$$n = 2^{a_0} p_1^{2a_1} \dots p_r^{2a_r} q_1^{b_1} \dots q_s^{b_s}$$

- The q_i are odd rational primes such that $q_i \equiv 1 \pmod{4}$. They decompose as $q_i = q_i \bar{q}_i$.
- The p_i are odd rational primes such that $p_i \equiv 3 \pmod{4}$. These remain inert in the Gaussian integers and must appear with even multiplicity. Their existence implies a common factor to all roots. Thus, we generally assume the norm is free of such primes.
- The factor of 2 contributes either a global scaling or a 45° rotation, neither of which changes the essential structure of the roots.

Any Gaussian integer $x = a + ib$ with norm $a^2 + b^2 = n$ satisfies (up to units):

$$x = \prod_{i=1}^s q_i^{e_i} \bar{q}_i^{b_i - e_i}$$

This is characterized by a vector $\mathbf{e} = (e_1, \dots, e_s)$ where $0 \leq e_i \leq b_i$.

For the sake of simplicity, in the next sections we only treat norms without multiplicities. This allows us to view $\mathbf{e}$ as a binary vector. In that setting, taking the logical NOT of the e_i is equivalent to taking the complex conjugate of q_i .

Joint Parameterization of n Norm-like Gaussian Integers

We show it is possible to parameterize any set of n Gaussian integers sharing the same norm using 2^n variables. We are interested in essentially different solutions, therefore we pay attention to common factors and the order of the roots.

Gaussian Integer Pairs and Application to the E_3 Problem

Let $x = a + ib, y = c + id$ be norm-like Gaussian integers associated with vectors $\mathbf{e}_x$ and $\mathbf{e}_y$. Let $g = m + in$ be the product of factors where $\mathbf{e}_x, \mathbf{e}_y$ are identical and, $h = p + iq$ where they differ.

$$\begin{aligned} x &= gh = mp - nq + i(np - mq) \\ y &= g\bar{h} = mp + nq + i(np + mq) \end{aligned}$$

This yields the Brahmagupta–Fibonacci parameterization of $a^2 + b^2 = c^2 + d^2$. Since the L_1 condition is the only constraint for the E_3 problem, every instance of this parameterization is a valid E_3 solution, as correctly pointed out in [3]. The converse also holds. Every E_3 solution can be parameterized by some g, h .

Gaussian Integer Quadruplets and Application to the E_4 Problem

To parameterize four norm-like Gaussian integers n_1, n_2, n_3, n_4 , we identify $2^3 = 8$ distinct “syndromes” (subsets of indices where factors are conjugated relative to n_1). Let $X_i = s_i + it_i$ be the product of all factors matching the i -th syndrome. Following this observation, we can write down the four Gaussian integers as:

$$\begin{aligned} n_1 &= A + iB = X_0 X_1 X_2 X_3 X_4 X_5 X_6 X_7 \\ n_2 &= C + iD = X_0 X_1 X_2 X_3 \overline{X_4 X_5 X_6 X_7} \\ n_3 &= E + iF = X_0 X_1 \overline{X_2 X_3} X_4 X_5 \overline{X_6 X_7} \\ n_4 &= G + iH = X_0 \overline{X_1 X_2 X_3} X_4 \overline{X_5 X_6 X_7} \end{aligned}$$

Each of the coefficients A, B, C, D, E, F, G, H can be thought of as a degree 8 polynomial in 16 variables $\mathbf{X} = \{s_i + it_i\}_{i=0}^7$.

The General Case

This generalizes to any number of integers. The exponential (2^{n-1}) growth of the number of possible syndromes rapidly makes the parameterizations hard to utilize. For e.g., in light of the E_5 problem, we can parameterize an octet of norm-like Gaussian integers. The expressions for the L_2 and L_3 conditions would result in dense polynomials with $2^8 = 256$ variables of degree no less than 512 and 1024 respectively.

Formulation of the E_4 Problem Using a Single Constraint

The L_2 condition can be written as (Here is the $L_{2,b}$ form, but the result is agnostic to this augmentation):

$$W(\mathbf{X}) := A(\mathbf{X})^2 B(\mathbf{X})^2 + C(\mathbf{X})^2 D(\mathbf{X})^2 - E(\mathbf{X})^2 F(\mathbf{X})^2 - G(\mathbf{X})^2 H(\mathbf{X})^2 = 0$$

By the construction of the parameterization, the L_1 condition:

$$\begin{aligned} A(\mathbf{X})^2 + B(\mathbf{X})^2 &= C(\mathbf{X})^2 + D(\mathbf{X})^2 \\ &= E(\mathbf{X})^2 + F(\mathbf{X})^2 = G(\mathbf{X})^2 + H(\mathbf{X})^2 \end{aligned}$$

is automatically satisfied. Therefore the E_4 problem now translates to finding the zeros of $W(\mathbf{X})$, a degree 32 polynomial in 16 variables. To the best of our knowledge, this is the first reduction of the E_4 problem into a single constraint problem (as opposed to the original formulation which involves 4 equations). Investigating $W(\mathbf{X})$ is a difficult task due to its size. The actual polynomial expression is far too big to be displayed here, and not even easily manipulated with a computer program. The following observation can be used to simplify the expression: Our quadruplet will not always contain all possible syndromes $X_0, \dots, X_7$, inviting us to address separately different “syndrome patterns” solutions might have. For e.g., a quadruplet resulting as:

$$\begin{aligned} n_1 &= X_0 X_1 X_4 X_7 \\ n_2 &= X_0 X_1 \overline{X_4} \overline{X_7} \\ n_3 &= X_0 X_1 X_4 \overline{X_7} \\ n_4 &= X_0 \overline{X_1} X_4 \overline{X_7} \\ X_2 &= X_3 = X_5 = X_6 = 1 + i0 \end{aligned}$$

Will be said to have a “ $\{0, 1, 4, 7\}$ ” syndrome pattern. It is therefore natural to ask the following questions:

- What syndrome patterns exist in the known families of [3] or in sporadic solutions.
- Are there some syndrome patterns that we can rule out altogether?
- Conversely, are some patterns more promising than others?

One should also note that in the context of the E_4 problem, the integers are divided into a pair of pairs:

$$\{\{n_1, n_2\}, \{n_3, n_4\}\}$$

And the L_2 condition is agnostic to the order within the pairs and between them. This imposes an equivalence on some of the syndrome patterns, inviting us to canonize the syndrome patterns with respect to this equivalence.

The $L_{2,d}$ Condition and Fourth Powers

The $L_{2,d}$ term corresponds to the real part of the fourth powers of the Gaussian integers:

$$L_{2,d}(a + ib, c + id) = \text{Re} \{ (a + ib)^4 + (c + id)^4 \}$$

Applying the Brahmagupta-Fibonacci parameterization:

$$\begin{aligned} L_{2,d}(gh, g\bar{h}) &= \text{Re} \{ (gh)^4 + (g\bar{h})^4 \} = \text{Re} \{ g^4(h^4 + \bar{h}^4) \} = 2\text{Re} \{ g^4 \} \text{Re} \{ h^4 \} \\ &= 2(m^2 - n^2 - 2mn)(m^2 - n^2 + 2mn)(p^2 - q^2 - 2pq)(p^2 - q^2 + 2pq) \end{aligned}$$

We believe that the factorization of the 8-th degree polynomial to four 2-nd degree components can be useful in both analytical and numerical studies of the search.

Single Constraint Formulation of E_4 - The $L_{2,d}$ Prism

The single-constraint formulation for E_4 becomes:

$$W(\mathbf{X}) = L_{2,d}(n_1(\mathbf{X}), n_2(\mathbf{X})) - L_{2,d}(n_3(\mathbf{X}), n_4(\mathbf{X})) = 0 \Rightarrow$$

$$\text{Re} \{ (X_0 X_1 X_2 X_3)^4 \} \text{Re} \{ (X_4 X_5 X_6 X_7)^4 \} = \text{Re} \{ (X_0 \bar{X}_3 X_4 \bar{X}_7)^4 \} \text{Re} \{ (X_1 \bar{X}_2 X_5 \bar{X}_6)^4 \}$$

This refactors the degree 32 equation into degree 8 terms, each involving only 8 variables, thus potentially simplifying numerical searches.

Implementation-Aware Enumeration of E_4 Solutions

Goal: Find all E_4 solutions for a given norm N .

Stage 1

- Factorize N and decompose $q_i \equiv 1 \pmod{4}$ into Gaussian factors.
- The $p_i \equiv 3 \pmod{4}$ primes and the factor of 2 only contribute global scaling or rotation so we omit them.

Stage 2

- Construct the set U of all Gaussian integers with norm $\tilde{N} = q_1^{b_1} \dots q_s^{b_s}$. We compute U in a tree-like fashion, branching over each factor and its conjugate.
 - Start with $U = \{1\}$
 - For each q_1 in $\tilde{N}$'s factorization (including multiplicities): $U \leftarrow q_1 U \cup \bar{q}_1 U$
- Canonize elements to the first octant ($a + ib : 0 \leq b \leq a$) and remove duplicates.
- Compute $r_{x,y} = (xy)^2$ (using two integer multiplications)

Stage 3

- Initialize a hash-table H .
- For each unordered pair $x + iy, u + iv \in U^2$
 - Compute key $k = r_{x,y} + r_{u,v}$
 - Try to insert $H[k]$, a collision yields a solution.

Complexity and Implementation Details

- Runtime and memory grow exponentially with the number of factors ($O(2^{2n})$).
- Large integer arithmetic can be optimized using reduced precision (e.g., $\mathbb{Z}/2^{32}\mathbb{Z}$) to detect collisions, followed by full-precision verification.
- In stage 2, We compute a total of 2^{n+1} gaussian integer multiplications, each such multiplication would normally cost 4 integer multiplications. We can utilize the fact that we also multiply with the conjugate to reduce this number to 2.
- $r_{x,y}$ was chosen with respect to $L_{2,b}$ and costs two multiplications. We could equivalently choose it to correspond to $L_{2,d}$ which would simply be the real component of the final integer, provided we precompute all factors to the fourth power (attractive in the reduced precision approach).
- The complexity of factorizing N is not discussed since in our context it will rarely be an arbitrary large random number.
- Computing the decompositions of the gaussian factors can be done as a preliminary stage and reused.
- We are likely to enumerate large sets of norms, therefore we should not find it hard to parallelize the search. Furthermore, jointly enumerating certain norms together allows us to amortize some of the work, for e.g., if the norms share factors that saves some of arithmetic in Stage 2.

Analysis of Known E_4 Solutions

We examined the following known solutions:

- The first family of [3] for $k = [1, 30]$.
- The second family of [3] for $k = [5, 60]$.
- Sporadic solutions 4 to 8 for table 1 of [3].
- A new solution found by our enumeration method.

For each solution we performed the following analysis:

- Canonized the solution:
 - Scaled to remove common factors.
 - Rotated to the first octant.
 - Transformed such that L_1 will be odd.
- Sorted the solution pairs in descending order.
- Computed L_1 and $L_{2,d}$. If they were not too large, we factorized them.

- Computed the canonized syndrome pattern by factoring each of the four Gaussian integers and analyzed factorizations.

For e.g., our new solution exhibits the following properties:

$$\begin{aligned}
A &= 252885 \quad B = 46703 \quad C = 195203 \quad D = 167415 \\
E &= 249703 \quad F = 61485 \quad G = 209985 \quad H = 148453 \\
L_1 &= 33065996717 = 13^1 \cdot 89^1 \cdot 173^1 \cdot 233^1 \cdot 709^1 \\
L_{2,d} &= 912832201535971887688 = -1 \cdot 2^3 \cdot 7^1 \cdot 17^1 \cdot 23^2 \cdot 31^1 \cdot 41^1 \cdot 47^1 \cdot 137^1 \cdot 401^1 \cdot 601^1 \cdot 919^1 \\
\text{Syndrome Pattern} &= \{0, 1, 3, 5\}
\end{aligned}$$

We can draw the following insights:

- None of the 90 unique solutions we examined shared a value of L_1 or $L_{2,d}$, thus failing to pair up to create an E_5 solution.
- Unique Syndrome patterns (calculated for the smallest 18 solutions) are:

$$\begin{aligned}
&\{0, 1, 5, 6\}, \{0, 1, 3, 5, 6\}, \{0, 1, 4, 6\} \\
&\{0, 1, 4, 5, 6\}, \{0, 1, 3, 4, 5, 6\}, \{0, 1, 2, 4, 5, 7\}, \\
&\{0, 1, 2, 4\}, \{0, 1, 3, 4, 6\}, \{0, 1, 3, 5\}, \{0, 1, 3, 4, 5\}
\end{aligned}$$

- The factorization of L_1 and $L_{2,d}$ appears to be surprisingly smooth (comparable to random integers of same size) and rarely contain multiplicities for odd primes.
 - Smoothness is probably related to the fact that under the parameterization, they are a product of 2 and 4 quadratic polynomials respectively.
- Furthermore, the primes composing $L_{2,d}$ are always $\equiv \pm 1 \pmod{8}$ (In L_1 they are $\equiv 1 \pmod{4}$ which aligns with L_1 being a norm of a Gaussian integer).

More Ideas

- Providing an estimate for the $E_{4/5}$ solution **density** to bound enumeration efforts.
- Systematically explore solution families via generalized linear constraints, extending the techniques of [3].
- Apply the Brahmagupta-Fibonacci parameterization to the $L_{2,b}$ or $L_{2,c}$ constraints, since they also can be viewed as sum of squares.
- The ring of Hurwitz quaternions possesses a norm which is a sum of four squares. We can use it to parameterize the expression for $L_{2,a}$, which is a sum of four fourth-powers.
- Apply a Chinese Remainder Theorem (CRT) approach: solve modulo various primes and lift to integers.

References

- [1] https://www.researchgate.net/publication/220975993_Few_Product_Gates_But_Many_Zeros
- [2] <https://mathworld.wolfram.com/SumofSquaresFunction.html>
- [3] <https://projecteuclid.org/journalArticle/Download?urlid=em%2F1243429952>